

Speech Tech Week 6 AQ PA GA

1) Which of the following is an output of a speech recognition block?

1 point

- ☒ text
- ☐ video
- ☐ audio
- ☐ image

speech → text

2) Excitation Signal is Periodic for Voiced Phonemes.

1 point

- ☒ true
- ☐ False

unvoiced
→ aperiodic

1) GMM is a discriminative algorithm.

1 point

- ☐ True
- ☒ False

2) Which of the following is the possibility of π_A , π_B and π_C with 3 mixture of gaussians?

1 point

- ☐ 0.1, 0.2, 0.6
- ☐ 0.3, 0.3, 0.3
- ☐ -0.4, 0.2, 1.2
- ☒ Neither of them

→ all of them
should add upto 1.

1) Which of the following is the most logical feature vector to build speech recognition classification model?

1 point

- ☐ Raw waveform
- ☐ Magnitude Spectrum
- ☐ Phase Spectrum
- ☒ MFCC

Log MFB

2) MFCC vector will be exactly similar for /aa/ sound corresponding to two different speaker.

1 point

- ☒ true
- ☐ False

their pitch, tone
might differ but
their MFCC will remain
same.

1) **Statement A:** Resonance frequency is likely to be higher of female as compared to male

1 point

Statement B: Vocal tract length of female is shorter than male

☒ Both A and B are correct

☐ Both A and B are incorrect

☐ A is correct and B is incorrect

☐ A is incorrect and B is correct

2) Given a Speech Recognition task of English audio, we have data of 20 Speakers and let's consider English contains approximately 40 phonemic sounds. Which of the following is more likely to select an option of number of mixtures to build GMM?

1 point

☐ 20

☒ 40

1) The word 'yes' can be broken into three phonemes /iy/, /eh/, and /s/. Which features are used for modeling GMM^{iy} , which is a statistical model for phoneme /iy/ trained over utterances of many individuals.

☐ Linear Features

☒ Log-Mel Filter Bank Features

☐ DFT Features

☐ Cepstrum Features

2) State True or False: For a word utterance, the phoneme wise GMM model is built only once

1 point

☐ True

☒ False

3) Speech is a seq. modelling problem and hence the use of HMM model is necessary
sequence

1) For a Hidden Markov Model of speech data phoneme is the hidden state and log MFB is the observed state.

1 point

☒ phonemes, Log-Mel Filter Bank features

☐ Log-Mel Filter Bank features, phoneme

☐ Cepstrum, Vocal Tract Excitation

☐ None of These

2) A Lexicon is a mapping from $\text{rule}(1\text{cm})(0.15\text{mm})$ to $\text{rule}(1\text{cm})(0.15\text{mm})$.

1 point

☐ music, sound

☐ phonemes, words

☒ words, phonemes

☐ utterance, word

1) What are resonant frequencies of Vocal Tract called?

1 point

- ☐ Pitch
- ☒ Formant
- ☐ Quefrequency
- ☐ None of the above

2) Mel frequency scale uses which of the following mathematical function?

1 point

- ☐ Antilog
- ☒ Logarithmic
- ☐ tanh
- ☐ sigmoid

log

3) Arrange Men, women and children corresponding to their length of vocal tract in ascending order.

1 point

- $C < W < M$

4) Number of phonemes will generally be higher than number of characters in a sentence.

1 point

- ☒ True
- ☐ False

mostly
yes

5) For a GMM model, covariance matrix estimation is tougher than Mean vector estimation because number of parameters in Covariance matrix are higher than Mean vector

1 point

- ☒ True
- ☐ False

$[=]_{2 \times 1}$
Mean

$[=]_{2 \times 2}$
Covariance

6) State True or False: $P(o_1/'ah')$ is the transition probability from 'ah' $\rightarrow$ o1. Here 'ah' is the phoneme, while o1 is the Log-Mel Filter Bank feature.

1 point

- ☐ True
- ☒ False

$\rightarrow$ observation
probability

7) State True or False: After uniformly segmenting a word utterance, and training phoneme wise GMM model. Labels of each feature is updated by comparing against the trained GMM model

1 point

- ☒ True
- ☐ False

1) Which of the following window should be used for windowing a signal?

1 point

☐ Rectangular

☒ Hamming

☒ Hanning

☒ both B and C

Yes, the answer is correct.

Score: 1

Accepted Answers:

both B and C

2) MFCC captures which of the following information?

1 point

☐ Vocal folds

☒ Vocal tract

☐ Both A and B

☐ Neither of them

3) Which of the following is the possibility of π_A , π_B and π_C with 3 mixture of gaussians, given that all examples are moreover same?

1 point

☐ 1, 0, 0

☐ 0, 1, 0

☐ 0, 0, 1

☒ All of them

Yes, the answer is correct.

Score: 1

Accepted Answers:

All of them

4) Classification of phonemes based on GMM model has high chances to get confused with which of the following phone?

1 point

☐ Right context phone

☐ Left context phone

☒ Both A and B

both
side confuses

far ← hear

5) Lets assume we have 3 sound /uu/, /ae/, /aa/. Which of the following two has high chances that their cluster will be nearby?

1 point

- ☒ /aa/ and /ae/
- ☐ /aa/ and /uu/
- ☐ /ae/ and /uu/

Yes, the answer is correct.

Score: 1

Accepted Answers:

/aa/ and /ae/

6) For the word 'how' state whether the following transition is allowed while using viterbi for searching best path: /h/ → /h/ → /ow/ → /h/ → /ow/

1 point

- ☐ Possible
- ☒ Not Possible
- ☐ Can't Say
- ☐ Need more information

not possible

7) For a Hidden Markov Model, we have two states: hidden and observed. Consider the weather forecast problem with the hidden states being: Sunny, Rainy and Cloudy. While the observed states being Dry, Dryish, Damp and Soggy. Given that previous day had cloudy weather. What is the probability that observed state today was Dry?

1 point

- ☐ $P(\text{Dry}/\text{Cloudy})$
- ☐ $P(\text{Dry})$
- ☐ $P(\text{Cloudy}/\text{Sunny})P(\text{Sunny}/\text{Dry}) + P(\text{Rainy}/\text{Dry})P(\text{Dry}/\text{Cloudy})$
- ☒ $P(\text{Dry}/\text{Sunny})P(\text{Sunny}/\text{Cloudy}) + P(\text{Dry}/\text{Rainy})P(\text{Rainy}/\text{Cloudy}) + P(\text{Dry}/\text{Cloudy})P(\text{Cloudy}/\text{Cloudy})$

Obs x Tran

Yes, the answer is correct.

Score: 1

Accepted Answers:

$P(\text{Dry}/\text{Sunny})P(\text{Sunny}/\text{Cloudy}) + P(\text{Dry}/\text{Rainy})P(\text{Rainy}/\text{Cloudy}) + P(\text{Dry}/\text{Cloudy})P(\text{Cloudy}/\text{Cloudy})$

8) Consider the word 'sand', we see that it can be split into 4 constituent phonemes /s/, /a/, /n/, and /d/. In how many parts would you divide the utterance of 'sand' for training phoneme wise GMM models.

1 point

- ☐ 1
- ☐ 8
- ☒ 4